THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x = 1, y = 3$ is a solution of $4x + 2y = k$, then k is: 1

(a) 14 **(b)** 10 **(c)** -2 **(d)** 24

2. The sum of two numbers is 18 and their difference is 2. The larger number is: 1

(a) 16 **(b)** 10 **(c)** 20 **(d)** 8

3. The graph of $x = 5$ is a line parallel to the: 1

(a) line $y = x$ **(b)** x-axis **(c)** y-axis **(d)** none of the axes

4. The lines $x + 3y = 6$ and $4x + 12y = 24$ are: 1

(a) intersecting at one point **(b)** coincident
(c) parallel **(d)** perpendicular

5. The pair of equations $x + 2y = 4$ and $3x + 6y = 8$ has: 1

(a) no solution (parallel lines)
(b) infinitely many solutions (coincident lines)
(c) exactly two solutions
(d) exactly one solution (intersecting lines)

6. Which of the following pairs (x, y) is a solution of $5x + y = 14$? 1

(a) $(4, 2)$ **(b)** $(-2, 4)$ **(c)** $(2, 4)$ **(d)** $(2, -4)$

7. If $x = 2, y = 2$ is a solution of $5x + 3y = k$, then k is: 1

(a) 16 **(b)** 4 **(c)** 60 **(d)** 32

8. The sum of two numbers is 30 and their difference is 12. The larger number is: 1

(a) 42 **(b)** 18 **(c)** 21 **(d)** 9

9. The pair of equations $2x - y = 3$ and $4x + y = 5$ has: 1

(a) no solution (parallel lines)
(b) infinitely many solutions (coincident lines)
(c) exactly two solutions
(d) exactly one solution (intersecting lines)

10. Which of the following pairs (x, y) is a solution of $4x + y = 9$? 1

(a) $(2, 1)$ **(b)** $(-2, 1)$ **(c)** $(2, -1)$ **(d)** $(1, 2)$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of k if $x = 1, y = 1$ is a solution of $2x + 7y = k$. 2

12. Check whether the pair $5x - 2y = 3$, $10x - 4y = 6$ is consistent. 2

13. Solve: $x + y = 30$, $x - y = 10$. 2

14. Find the value of k if $x = 2, y = -1$ is a solution of $5x + 3y = k$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by elimination: $x + 3y = 12$ and $2x + y = 14$. 3

16. Solve by the substitution method: $2x + y = 10$ and $x - y = 2$. 3

17. The sum of two numbers is 41 and their difference is 13. Find the numbers. 3

OR

Solve by elimination: $2x + 3y = 18$ and $x + y = 7$. 3

18. Solve by the substitution method: $x + y = 8$ and $x - y = 2$. 3

OR

The sum of two numbers is 53 and their difference is 37. Find the numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Solve by elimination: $2x - 3y = 6$ and $3x + 2y = 22$. 5

OR

5 pencils and 2 erasers cost ₹38, while 2 pencils and 3 erasers cost ₹35. Find the cost of one pencil and one eraser. 5

20. Solve by elimination: $3x + 4y = 20$ and $5x - 2y = 16$. 5

OR

3 pencils and 4 erasers cost ₹51, while 4 pencils and 1 erasers cost ₹42. Find the cost of one pencil and one eraser. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND